

TOURNAMENT AUCTION SYSTEM

Platform Workflow & Architecture Guide

1. Executive Overview

TournamentAuctionX is a state-of-the-art live player auction platform designed for sports, gaming, and competitive leagues. It facilitates team creation, player registration, admin review, and real-time bidding using socket-based connectivity. This document covers the complete workflow of the system from initial configuration to team finalization.

2. System Roles & Responsibilities

Administrator (Admin)

Sets up tournaments, approves or rejects player profile requests, configures teams, promotes registered players to team captains, and starts/stops/pauses the live auction.

Registered Player

Registers an account, uploads a profile photo, inputs player stats/experience, and tracks approval and auction results. Once approved, the player enters the random auction pool.

Team Captain (Bidder)

Appointed from the pool of registered players by the admin. Captains manage their team budget, bid on active players in real-time, and build their squad up to the tournament limit.

3. Step-by-Step Workflow

1. Phase I: Tournament Setup & Initialization

The Administrator creates a new tournament in Draft mode, specifying Name, Location, Date, Points per Team (budget), Squad Size Limits, and Bidding Timer duration. Once finalized, the admin activates the tournament, opening up player registrations.

2. Phase II: Player Registration & Photo Upload

Athletes sign up for player accounts and access the registration portal. They enter details (age, position, experience, foot preference, skill rating) and upload a profile photo. The system reads the photo file, encodes it as a Base64 string, and saves it in the database. Registrations enter a "Pending" queue.

3. Phase III: Admin Review & Team Assignment

The Administrator approves or rejects pending registrations. Approved players are queued for auction. The Admin creates teams, assigns an optional team mascot/icon, and selects the captain for each team from the pool of registered players. When selected, the player is promoted to a captain user account, assigned to captain their team, and automatically added to the team squad at a cost of 0 points.

4. Phase IV: Live Socket Bidding Room

The Admin launches the auction room. The system picks a random approved player and displays their photo and stats on the screen. The bidding timer starts. Team captains place bids in real-time using socket.io. If a bid is placed in the final 10 seconds, the timer resets to 10 seconds (sniping protection). The team with the highest bid when the timer reaches 0 wins the player.

5. Phase V: Squad Finalization & Completion

When a player is sold, points are deducted from the winning team budget, and the player is assigned to their squad. If no bids are placed, the player is marked unsold. The process repeats until all approved players are auctioned or the admin ends the session. The tournament is then marked "Completed".

4. Technical Specifications & Business Rules

Bidding & Budget Integrity Constraints

- A captain cannot bid an amount exceeding their team's remaining points balance.
- A captain cannot place a bid that is equal to or lower than the active highest bid.
- A captain cannot bid if their team squad has already reached the maximum squad limit.
- Bids must be placed in increments of 10 points (adjustable depending on tournament scale).
- Captains who are appointed directly by the admin are added to the squad with a bid price of 0, bypassing the auction pool.
- Profile images are stored natively as Base64 strings, eliminating dependencies on external file storage services.

Technical Architecture Overview

The system is built on a modern decoupled client-server architecture:

- Frontend: React.js application with Vite tooling, Tailwind CSS for styling, and Socket.io-client for real-time bidirectional messaging.
- Backend: Node.js Express server handling REST endpoints and authentication (JWT). Node-HTTP server handles socket rooms corresponding to active tournaments.
- Database: Relational schema using MySQL for absolute data integrity, locking budget variables, and registering bid timestamps.
- Storage: Base64 data encoding for high-speed file storage in the relational database, avoiding external storage service latency.

TournamentAuctionX Certification

This workflow document complies with the implementation specifications approved on June 1, 2026. All backend REST endpoints, Socket.io controllers, and frontend layout rendering modules have been aligned with this specification to ensure atomic operations and visual layout consistency.